HOMEWORK EXERCISE 5 SOLUTIONS

1) STEP 1: $H_0: \mu = 60$ [or $\mu \geq 60$]

$H_a: \mu < 60$

STEP 2: $\alpha = 0.01$

STEP 3: USE WILCOXON SIGNED RANK TEST W

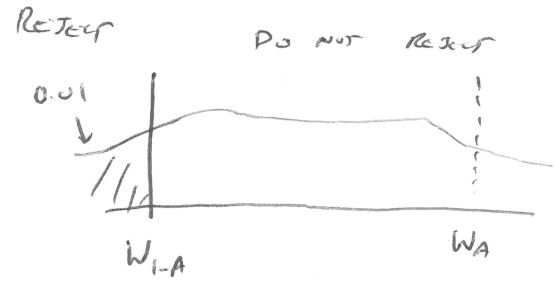
DATA COME FROM SYMMETRIC DISTRIBUTION, PLUS, WE WERE TOLD TO :)

STEP 4:

S1: H_a: "<" → LEFT TAIL TEST

S2: $\alpha = 0.01$

S3: W → USE TABLE V



$n = 17$, BUT ONE OF THE SAMPLE VALUES $= 60$ (H_0)

SO $n' = 16$, $\alpha = 0.01$ $\Big\}$ $W_A = 112$ FROM TABLE V

BUT THIS WOULD BE RIGHT-SIDE VALUE!

SO FIND $W_{1-A} = \dfrac{n'(n'+1)}{2} - W_A$

$W_{1-A} = \dfrac{16(17)}{2} - 112$

$W_{1-A} = \underline{\underline{24}}$

DECISION RULE: REJECT H_0 IF $W_{CALC} < 24$

STEP 5

DATA (x)	diff x − 60	|diff|	RANK |diff|	SIGN RANK
54	−6	6	5	−5
23	−37	37	16	−16
47	−13	13	9	−9
59	−1	1	1	−1
35	−25	25**	11.5	−11.5
53	−7	7	6	−6
49	−11	11	8	−8
35	−25	25**	11.5	−11.5
60	0	0	—	—
57	−3	3*	3	−3
63	3	3*	3	$\boxed{+3}$
57	−3	3*	3	−3
89	29	29	14	$\boxed{+14}$
50	−10	10	7	−7
33	−27	27	13	−13
28	−32	32	15	−15
41	−19	19	10	−10

W_{CALC} = SUM OF POSITIVE RANKS

* TIE WITH PROPOSED RANKS 2, 3, 4 → AVG RANK = $\frac{2+3+4}{3} = 3$

** TIE WITH PROPOSED RANKS 11, 12 → AVG RANK = $\frac{11+12}{2} = 11.5$

$W_{CALC} = 3 + 14 = 17$

SINCE $W_{CALC} = 17 < 24$, WE REJECT H_0.

THERE IS SUFFICIENT EVIDENCE TO SUGGEST THE MEAN RESPONSE TIME FOR THIS TYPE OF DISK DRIVE IS LESS THAN 60 MS.

2) a) THE SAMPLES ARE INDEPENDENT; SAMPLE STANDARD DEVIATIONS ARE KNOWN, BUT POPULATION STANDARD DEVIATIONS ARE NOT.

b) ANSWER (V): TWO SAMPLE t - HYPOTHESIS TEST.